

Information Acquisition and the Finance-Uncertainty Trap

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*The views are those of the authors and do not necessarily reflect the views of the Federal Reserve Bank of San Francisco or the Federal Reserve System.

This paper

- ▶ Studies the interaction b/w information acquisition (IA) and credit constraints
 - ▶ Theory: tractable heterogeneous-firm GE model
 - ▶ Empirics: novel measures of firm-level IA and evidence on two-way feedback

This paper

- ▶ Studies the interaction b/w **information acquisition (IA)** and **credit constraints**
 - ▶ Theory: tractable heterogeneous-firm GE model
 - ▶ Empirics: novel measures of firm-level IA and evidence on two-way feedback
- ▶ GE model features endogenous IA and endogenous credit constraints
 - ▶ IA depends on the **option value** of acquiring a more precise signal
 - ▶ Forward-looking credit constraint derived from the **IC condition** of creditors
- ▶ Two-way feedback between IA and credit constraints
 - ▶ Credit constraints reduce the option value of precise signals, discouraging IA
 - ▶ Discouraged IA tightens credit screening, amplified by GE channels

Information frictions and financial frictions reinforce each other, creating a **finance-uncertainty trap**.

Empirical evidence

▶ Two measures of IA

- ▶ *Input-based*: IA-job postings (Lightcast), IA workers (Revelio Labs) Measure
- ▶ *Outcome-based*: forecast precision in managerial guidance (I/B/E/S)
- ▶ Validated by input-outcome linkage, event studies, industry patterns Validation
- ▶ Robust to alternative occupation taxonomies and skill-based classifications

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▶ Credit tightening discourages IA

- ▶ IA-related job (posting) shares ↓ and managerial FE ↑ after credit tightening LP
- ▶ Stronger effects for firms with higher external finance dependence

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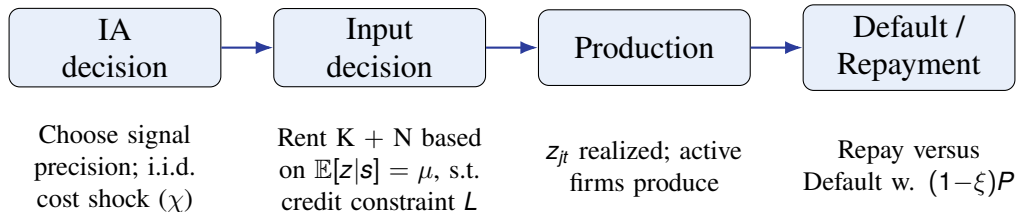
- ▶ IA-related job (posting) shares ↓ and managerial FE ↑ after credit tightening LP
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▶ Exogenous IA-cost shocks tighten credit access

- ▶ Exogenous changes in financial news supply driven by Ad. ↓ (Hu, 2024)
- ▶ Leave-one-out design: info supply shock orthogonal to info demand IV
- ▶ Credit access declines: debt and equity issuance ↓, cost of capital ↑ Results
- ▶ Channel operates through managerial information quality

Data suggests a two-way feedback loop between IA and credit access.

Model: a summary



- ▶ Hetero. firms with CRS technology; idios. productivity z_{jt} unknown ex ante
- ▶ Information friction: firms must hire χ_{jt} workers to improve signal precision
- ▶ Financial friction: working capital $RK + WN$ must be financed externally
- ▶ Default risk + info asymmetry $\rightarrow$ endogenous credit limit L
- ▶ All idiosyncratic shocks (z, χ) are i.i.d. $\rightarrow$ exact aggregation

Input decisions

- ▶ IC (no default) condition of creditors → cash-flow-based credit constraint

$$L_{j,t} \leq \xi_t \underbrace{\mathbb{E}_t M_{t+1} V_{j,t+1}}_{\text{discounted cash flow}}$$

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- ▶ *CRS*+ i.i.d. $z \rightarrow$ A firm is active iff its expected productivity $\mu_{j,t} \equiv \mathbb{E}_t[z_{j,t}|\mathbf{s}_{j,t}]$ is high

$$k_{j,t} = \begin{cases} \frac{\alpha}{R_t} L_{j,t}, & \mu_{j,t} \geq \mu_t^* \\ 0, & \mu_{j,t} < \mu_t^* \end{cases} \quad n_{j,t} = \begin{cases} \frac{1-\alpha}{W_t} L_{j,t}, & \mu_{j,t} \geq \mu_t^* \\ 0, & \mu_{j,t} < \mu_t^* \end{cases}$$

where the **threshold** μ_t^* increases with factor prices W_t, R_t

Information decision

- ▶ Trade-off of IA
 - ▶ Cost: i.i.d. labor cost $\chi_{j,t}$
 - ▶ Benefit: option value of precise signals, profit rate $\pi^I(\mu_t^*) > \pi^U(\mu_t^*)$
- ▶ Lemma 3: firm acquires information iff it draws χ_{jt} below a threshold χ_t^*

$$\chi_t^* \cdot W_t = \underbrace{[\pi^I(\mu_t^*) - \pi^U(\mu_t^*)]}_{\Delta\pi: \text{option value}} L_t$$

- ▶ Prop 1: Tightening credit constraints reduces IA: $L_t \downarrow \rightarrow \chi_t^* \downarrow$

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- ▶ Prop 1: **Tightening credit constraints reduces IA**: $L_t \downarrow \rightarrow \chi_t^* \downarrow$
- ▶ The equilibrium is **fully characterized by two endogenous cutoffs**: μ_t^* and χ_t^*
 - ▶ Firms with $\chi_{j,t} \leq \chi_t^*$ acquire information
 - ▶ Firms with $\mu_{j,t} \geq \mu_t^*$ hire inputs and produce

Credit constraint

- ▶ Loan capacity $\approx$ firm continuation value (*average quality of active firms*)

$$L_t \approx \xi P_t(\pi)$$

where average profit rate π increases with the share (prob.) of informed firms

- ▶ Prop 2: **IA increases credit capacity**: $\chi^* \uparrow \rightarrow \pi \uparrow \rightarrow L \uparrow$

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- ▶ Prop 2: **IA increases credit capacity**: $\chi^* \uparrow \rightarrow \pi \uparrow \rightarrow L \uparrow$
- ▶ Prop 3 (**Spillover**): IA by firm $i \rightarrow$ benefit of IA by firm $j \uparrow$
 - ▶ Credit reallocation (*GE effect*): $W \uparrow \Rightarrow \underbrace{1 - F(\mu^*)}_{\text{selection}} \downarrow \cdot \underbrace{L/W}_{\eta_{jt}} \uparrow \uparrow \Rightarrow \underbrace{(\pi^I - \pi^U)}_{\text{option value}} \uparrow$
- ▶ Prop 4 (**Dynamic feedback**): Info. $\chi_{t+1}^* \uparrow \rightarrow P_t \uparrow \rightarrow$ credit $L_t \uparrow \rightarrow$ info. $\chi_t^* \uparrow$
 - ▶ Prediction: negative news about firm future cash flows would depress IA
 - ▶ Supported by evidence based on oil supply news shock (Känzig, 2021) Evidence

Aggregation & Calibration

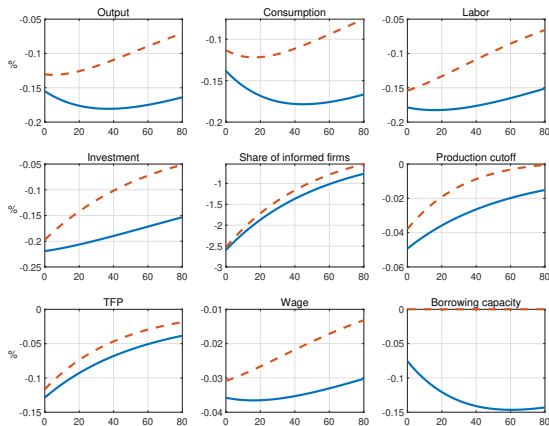
- ▶ No distribution to track → closed-form aggregation:

$$TFP = \frac{F(\chi_t^*)[1 - G^I(\mu_t^*)]A_t^I + [1 - F(\chi_t^*)][1 - G^U(\mu_t^*)]A_t^U}{\underbrace{F(\chi_t^*)[1 - G^I(\mu_t^*)]}_{\text{Mass: active informed firms}} + \underbrace{[1 - F(\chi_t^*)][1 - G^U(\mu_t^*)]}_{\text{Mass: active uninformed firms}}}.$$

- ▶ Solved by **third-order perturbation** ⇒ captures nonlinear risk effects
- ▶ Five micro moments identify information-friction parameters in the model

Moment	Data	Model	Identifies
SD of log markup	0.30	0.30	Prior uncertainty about productivity
Rel. SD of log forecast error	0.91	0.90	Average signal precision
SD of log FE: informed / uninformed	0.70	0.70	Precision gain from acquiring info
Share of informed firms	0.50	0.50	Dispersion of information costs
Average IA-job posting share	0.07	0.07	Average level of information cost

Credit constraints amplify information shocks



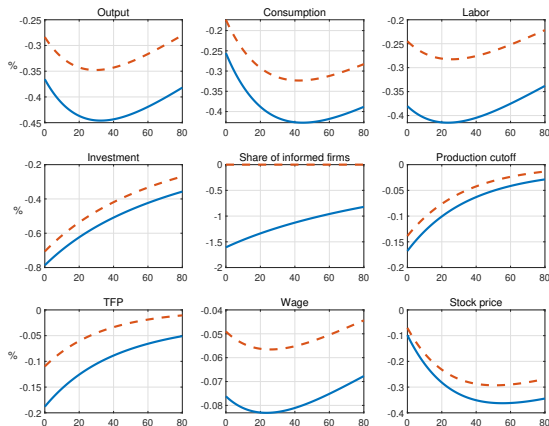
Experiment: raise the average cost of information acquisition by 1%.

Counterfactual: hold borrowing capacity fixed.

Endogenous credit makes info-cost shocks more contractionary and persistent.

Info-cost shocks $\Rightarrow$ higher uncertainty and lower equity $\Rightarrow$ tighter credit that amplifies the recession.

Information acquisition amplifies credit shocks



Counterfactual: hold IA fixed.

Endogenous IA magnifies output and TFP effects of credit tightening: the peak output decline is $\approx 28\%$ larger than in the counterfactual.

Credit shocks $\Rightarrow$ lower option value of IA $\Rightarrow$ less learning, lower TFP and cash flows.

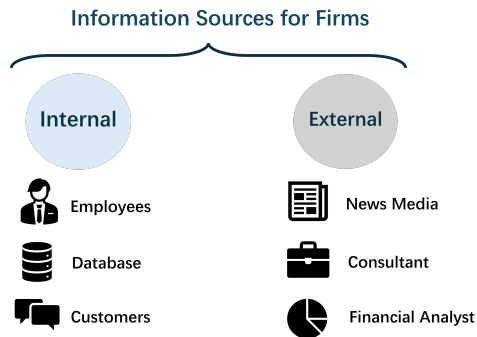
Conclusion

- ▶ Information acquisition is a measurable firm investment (job-posting share).
- ▶ Information-cost shocks causally tighten financing by raising managerial uncertainty.
- ▶ Feedback between information and credit creates a quantitatively important *finance-uncertainty trap*.
- ▶ Not for today but in the paper:
 - ▶ Internal + external IA sources in the GE model
 - ▶ Policy interventions to restore the second-best

Measuring and validating firm-level IA

- ▶ Online job postings from **Lightcast** (near-universe since 2010).
- ▶ IA jobs: gather and analyze data relevant for profitability / productivity (LLM + human audit).
- ▶ Measure: **IA-job share** = IA postings / all postings (firm-quarter).

Occupations



Information-related occupations

1. Business Analysis and Strategy

Business Analyst, Strategic Planner/Analyst, Economist, Survey Researcher, . . .

2. Operation and Supply Chain Management

Operation Research Analyst, Production Planner, Logistician, Supply Chain Specialist, . . .

3. Financial and Investment Management

Financial Analyst, Investment/Portfolio Analyst, Budgeting Analyst, Risk Manager, . . .

4. Marketing and Sales

Brand Manager, Marketing Manager, Digital Marketing Analyst, Marketing Automation Manager, . . .

5. Data Analytics and Business Intelligence

Data Analyst, Data Scientist, Data Manager, . . .

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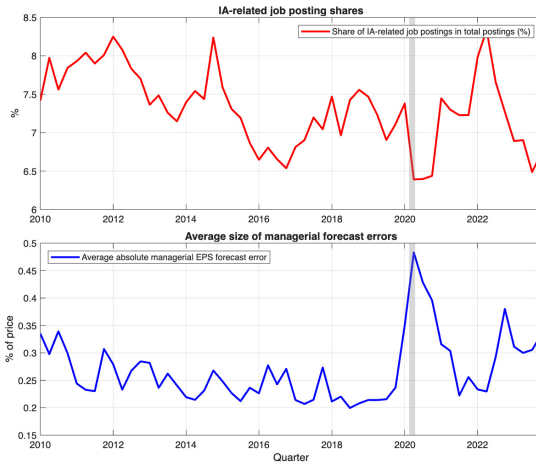
Validating our measure of IA

	Uninformed	Informed
IA-job share (%)	4.01	14.52
Managerial FE ($\times 100$)	0.24	0.21
LW constraint index	0.15	0.04

- ▶ Informed firms: IA-job share above industry $\times$ quarter median.
- ▶ Smaller managerial forecast errors (I/B/E/S) and easier credit access (Linn–Weagley).

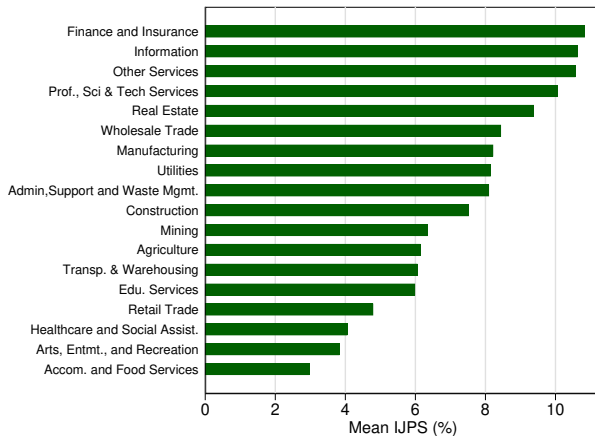
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Average share of info-related job postings across time

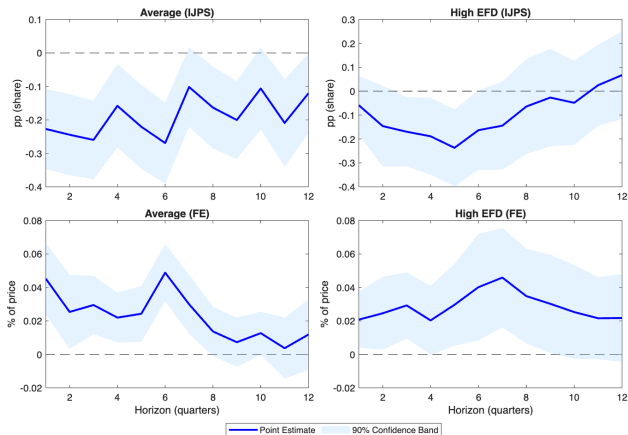


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Average share of info-related job postings across sectors



Credit tightening reduces information acquisition



► GZ tightening

⇒

↓ in IA-job share

↑ in managerial FE

► Stronger for firms in high external-finance-dependence industries.

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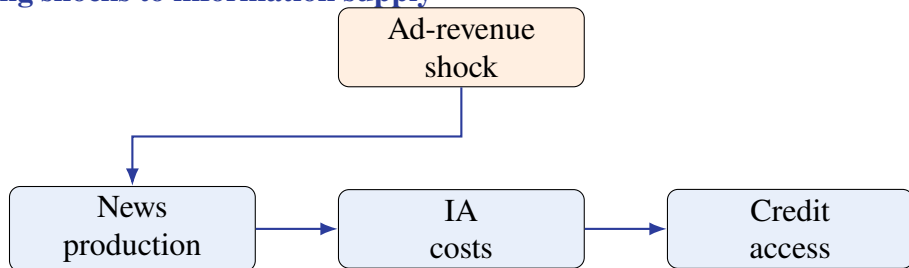
External and internal information are imperfect substitutes

Dependent variable	$IJPS_{j,t+1}$	$\Delta \log(FE_{j,t}^F)$
$\Delta FinNews_{j,t}$	-0.389^{**} (0.154)	-0.139^{***} (0.052)
Firm controls, firm FE, time FE	Yes	Yes

Less external news raises internal IA hiring but still worsens managerial forecast precision.

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Identifying shocks to information supply



- ▶ Financial media rely on advertising and subscription revenues (Hu, 2024).
- ▶ Leave-out ad shocks reduce news production for covered firms $\Rightarrow$ raise effective IA costs.
- ▶ Leave-one-out exposure excludes the firm's own industry (newsworthiness / demand concerns).

Information shocks tighten financing conditions

Dependent variable	Δ LT debt	Δ equity	log cost of capital
$\Delta FinNews$	0.229** (0.103)	0.552*** (0.097)	-0.007*** (0.002)
Firm controls, firm FE, time FE	Yes	Yes	Yes

Less media coverage



Managerial FE $\uparrow$



External financing $\downarrow$

- ▶ Declines in news reduce LT debt and equity issuance and raise CoC.
- ▶ Channel: higher managerial forecast errors (survives analyst FE controls).

Exogenous IA-cost shocks causally tighten firms' access to external finance.

Channel: fitted managerial FE and financing

	$\Delta Debt^{ST}$	$\Delta Debt^{LT}$	$\Delta Equity$	$\log(CoC)$
$\Delta \widehat{\log(FE^F)}$	0.007 (0.207)	-0.701** (0.274)	-0.432*** (0.143)	0.038*** (0.008)
$\log(FE^A)$	-0.008	0.017**	-0.005	0.001***
Firm FE, time FE, controls	Yes	Yes	Yes	Yes

News-driven rises in managerial FE line up with tighter external finance after controlling for analyst FE.

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A GE framework: key ingredients

- ▶ CRS technology $y_{jt} = z_{jt} k_{jt}^{\alpha} n_{jt}^{1-\alpha}$; idiosyncratic z_{jt} unknown ex ante
 - ▶ Idiosyncratic productivity $z_{jt} \sim \mathcal{N}(-\frac{1}{2}\sigma_z^2, \sigma_z^2)$ unknown at the beginning
- ▶ Stage I: noisy signal $s_{jt} = \log(z_{jt}) + e_{jt}$; firm can raise precision $\tau \in \{\tau^l, \tau^u\}$ by hiring IA labor at cost χ_{jt}
 - ▶ External signal s_{jt}^{EX} with time-varying precision τ_t^{EX} captures news supply
- ▶ Stage II: inputs based on $\mu_{jt} \equiv \mathbb{E}[z_{jt}|s_{jt}]$ s.t. working-capital constraint $R_t k_{jt} + W_t n_{jt} \leq L_{jt}$
- ▶ Stage III: productivity realizes; repay or default; lender seizes fraction ξ_t of continuation value $\Rightarrow$ value-based credit constraint

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Supporting evidence for model mechanism

- ▶ Model predicts that a **news shock** reducing firm future cash flows would depress current stock price and hinder info acquisition
- ▶ Model predictions supported by evidence from oil supply news shocks
 - ▶ Oil supply news shock identified from OPEC production announcements, with delayed implementation for 6–24 months (Känzig, 2021)
- ▶ Findings: Positive oil supply news
 - ▶ Reduces expected future oil prices
 - ▶ Lowers current stock prices of exposed firms (i.e., those with positive oil beta)
 - ▶ Discourages IA efforts of exposed firms
 - ▶ Effects on IA work through credit channel (LW)